DataBase Management System

Library Management System



Presented by
-Pavani Venkata Sai

Project Overview

A database-driven system for organizing common library operations

Theme

A Library Management System maintains book details, member information, supplier records, book issue/return transactions, and fine details in a structured database.

Centralized records

Store library information consistently in related tables.

Faster retrieval

Use SQL queries to find members, books and issue details.

Better tracking

Track return status, due dates and fine ranges.

Problem Statement & Objectives

Why a structured library database is useful

Problem statement

Manual library records can be difficult to search, update and maintain. Book issue information, member details and fines may become inconsistent.

Project objectives

- Maintain member and book records
- Track book issues and returns
- Store supplier and fine information
- Retrieve information with SQL queries
- Maintain data consistency with keys and relationships

Keys & Constraints

How the database maintains data quality

Primary key (PK)

Uniquely identifies each row, e.g. MEMBER_ID or BOOK_CODE.

Foreign key (FK)

Connects related tables and helps prevent invalid references.

Data types

VARCHAR for text, DATE for dates, DECIMAL for prices.

Relationships

- Book issues reference members and books.
- Books reference suppliers.
- Book issues reference fine ranges.

Concepts Implemented

01 Entity-Relationship (ER) Modeling to design entities, attributes, and relationships.

02 Primary & Foreign Key constraints to enforce unique identity and referential integrity

03 Relational Schema Design with normalized master and transactional tables.

04 Data Definition Language (DDL) to create tables with proper data types and constraints.

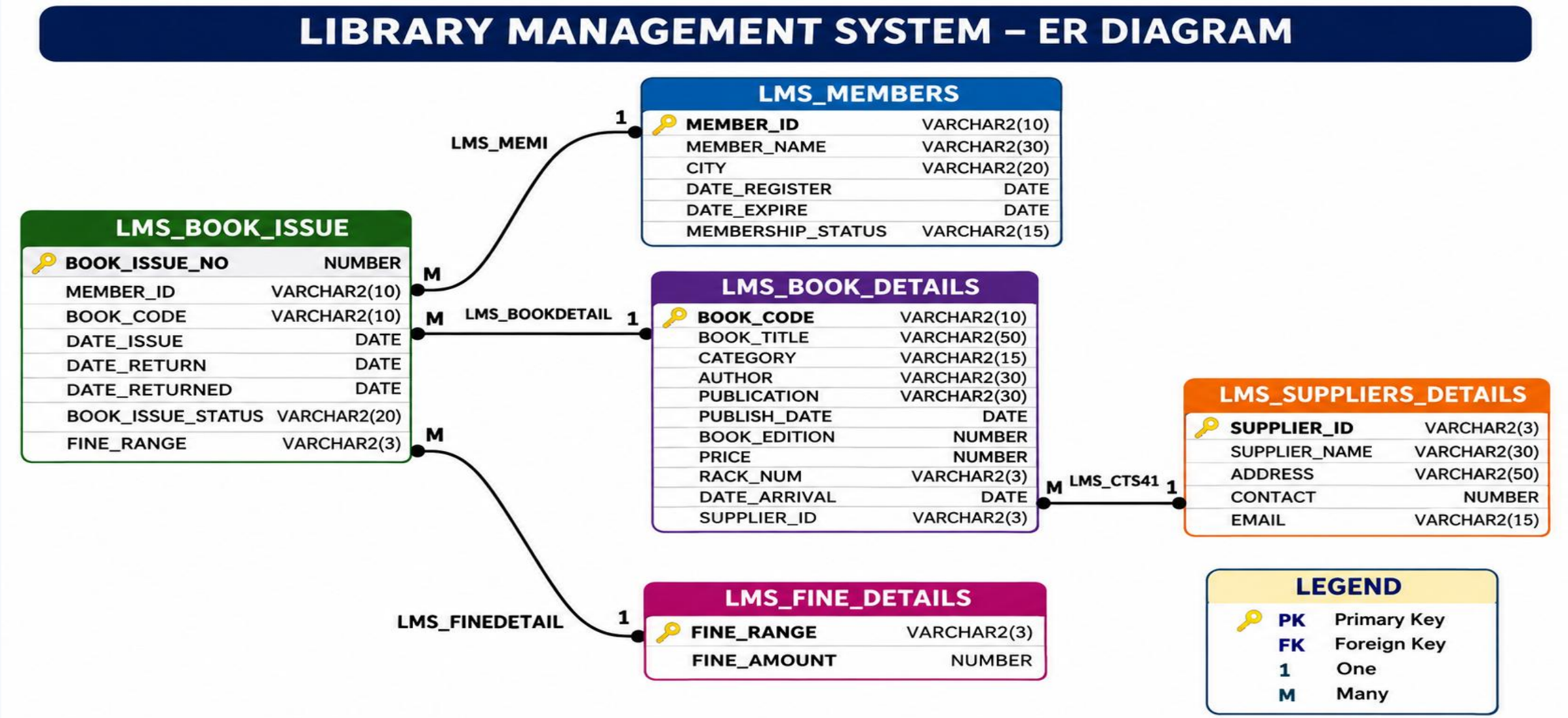
05 Data Manipulation Language (DML) to insert and manage sample records

06 Aggregate Functions & GROUP BY, including COUNT, AVG, and MAX, for summarized reporting.

07 Multi-Table JOIN Queries to combine related tables for connected, meaningful data.

Entity–Relationship Diagram

Tables and relationships based on the supplied LMS ER diagram



Database Tables

Five related tables form the core of the system

LMS_MEMBERS

Member ID, name, city, registration date, expiry date, membership status

LMS_BOOK_DETAILS

Book code, title, category, author, publication, dates, edition, price, rack and supplier

LMS_BOOK_ISSUE

Issue number, member ID, book code, issue/return dates, status and fine range

LMS_SUPPLIERS_DETAILS

Supplier ID, supplier name, address, contact and email

LMS_FINE_DETAILS

Fine range and fine amount

Sample Data — LMS Members, LMS Suppliers details

LMS Members

MEMBER_ID	MEMBER_NAME	CITY	DATE_REGISTER	DATE_EXPIRE	MEMBERSHIP_STATUS
M001	Ravi Kumar	Hyderabad	2011-01-10	2012-12-31	Permanent
M002	Sita Devi	Vijayawada	2011-02-15	2013-02-28	Temporary
M003	Arjun Rao	Eluru	2011-03-01	2012-08-31	Permanent
M004	Priya Sharma	Chennai	2012-01-10	2014-12-31	Temporary
M005	Kiran Kumar	Bengaluru	2011-05-05	2012-11-30	Permanent
M006	Lakshmi Devi	Guntur	2012-02-12	2015-06-30	Temporary
M007	Rahul Verma	Delhi	2011-06-01	2013-03-31	Permanent
M008	Anusha Reddy	Nellore	2012-04-15	2014-04-30	Temporary
M009	Vamsi Krishna	Ongole	2011-08-10	2012-10-31	Permanent
M010	Sneha Patel	Mumbai	2012-05-20	2015-12-31	Temporary
M011	Pavan Kumar	Kakinada	2011-09-01	2012-09-30	Permanent
M012	Divya Sri	Tirupati	2012-02-15	2016-06-30	Temporary
M013	Suresh Babu	Warangal	2011-11-01	2013-01-31	Permanent
M014	Meena Kumari	Rajahmun...	2012-06-10	2014-06-30	Temporary
M015	Sai Charan	Bhimavaram	2011-12-01	2012-07-31	Permanent
M016	Pooja Singh	Pune	2012-02-20	2013-03-15	Temporary
M017	Naveen Reddy	Kadapa	2011-07-01	2012-06-30	Permanent
M018	Harika Rao	Vizag	2012-03-15	2015-03-31	Temporary
M019	Manoj Kumar	Kurnool	2011-10-01	2013-02-15	Permanent
M020	Keerthi Reddy	Eluru	2012-02-25	2012-12-31	Temporary
NULL	NULL	NULL	NULL	NULL	NULL

LMS Suppliers details

SUPPLIER_ID	SUPPLIER_NAME	ADDRESS	CONTACT	EMAIL
S01	Prentice Hall	New Delhi	9876500001	s01@gmail.com
S02	Pearson	Hyderabad	9876500002	s02@gmail.com
S03	McGraw Hill	Chennai	9876500003	s03@gmail.com
S04	Wiley	Mumbai	9876500004	s04@gmail.com
S05	Oxford Press	Bengaluru	9876500005	s05@gmail.com
S06	Tata McGraw	Vijayawada	9876500006	s06@gmail.com
S07	S Chand	Delhi	9876500007	s07@gmail.com
S08	Dreamtech	Pune	9876500008	s08@gmail.com
S09	PHI Learning	Kolkata	9876500009	s09@gmail.com
S10	BPB Publications	Hyderabad	9876500010	s10@gmail.com
S11	Tech Books	Eluru	9876500011	s11@gmail.com
S12	Global Books	Guntur	9876500012	s12@gmail.com
S13	National Publish...	Chennai	9876500013	s13@gmail.com
S14	Classic Books	Mumbai	9876500014	s14@gmail.com
S15	Bright Publications	Pune	9876500015	s15@gmail.com
S16	Modern Books	Delhi	9876500016	s16@gmail.com
S17	Knowledge Publ...	Vizag	9876500017	s17@gmail.com
S18	Universal Books	Nellore	9876500018	s18@gmail.com
S19	Star Publishers	Kurnool	9876500019	s19@gmail.com
S20	City Publications	Eluru	9876500020	s20@gmail.com
NULL	NULL	NULL	NULL	NULL

Sample Data — LMS Fine details, LMS Book details

LMS Fine details

FINE_RANGE	FINE_AMOUNT
F01	0.00
F02	5.00
F03	10.00
F04	15.00
F05	20.00
F06	25.00
F07	30.00
F08	35.00
F09	40.00
F10	45.00
F11	50.00
F12	55.00
F13	60.00
F14	65.00
F15	70.00
F16	75.00
F17	80.00
F18	85.00
F19	90.00
F20	100.00
NULL	NULL

LMS Book details

BOOK_CODE	BOOK_TITLE	CATEGORY	AUTHOR	PUBLICATION	PUBLISH_DATE	BOOK_EDITION	PRICE	RACK_NUM	DATE_ARRIVAL	SUPPLIER_ID
BL000001	Database Management Systems	Education	Rajiv Chopra	Prentice Hall	2011-01-10	3	550.00	R01	2012-01-10	S01
BL000002	Java Programming	Java	Amit Kumar	Prentice Hall	2011-02-15	2	450.00	R02	2012-01-15	S01
BL000003	Core Java	Java	Sandeep Kumar	Pearson	2011-03-20	5	600.00	R03	2012-02-01	S02
BL000004	Advanced Java	Java	Praveen Kumar	Prentice Hall	2011-04-10	2	650.00	R04	2012-02-10	S01
BL000005	Python Programming	Programming	Rahul Sharma	Wiley	2011-05-12	3	500.00	R05	2012-03-05	S04
BL000006	C Programming	Programming	Vijay Kumar	Prentice Hall	2011-06-18	4	400.00	R06	2012-03-12	S01
BL000007	Computer Networks	Education	Anil Kumar	Pearson	2011-07-20	5	600.00	R07	2012-04-01	S02
BL000008	Operating Systems	Education	Ramesh Babu	McGraw Hill	2011-08-25	4	650.00	R08	2012-04-15	S03
BL000009	Data Structures	Programming	Kiran Reddy	Prentice Hall	2011-09-15	3	580.00	R09	2012-05-10	S01
BL000010	Web Development	Technology	Pavan Kumar	Pearson	2011-10-10	2	500.00	R10	2012-05-20	S02
BL000011	Java Fundamentals	Java	Arun Kumar	Prentice Hall	2011-11-05	6	700.00	R11	2012-06-01	S01
BL000012	Software Engineering	Education	Srinivas Rao	McGraw Hill	2011-12-01	7	720.00	R12	2012-06-15	S03
BL000013	Machine Learning	Technology	Priya Sharma	Wiley	2012-01-15	2	750.00	R13	2012-07-01	S04
BL000014	Computer Architecture	Education	Ravi Shankar	Prentice Hall	2012-02-20	4	680.00	R14	2012-07-10	S01
BL000015	Cloud Computing	Technology	Sai Krishna	Pearson	2012-03-25	3	620.00	R15	2012-08-01	S02
BL000016	Java Complete Reference	Java	Venkat Subram...	Prentice Hall	2012-04-12	8	800.00	R16	2012-08-15	S01
BL000017	Cyber Security	Technology	Pradeep Kumar	Wiley	2012-05-18	2	690.00	R17	2012-09-01	S04
BL000018	Algorithms	Education	Deepak Gupta	Prentice Hall	2012-06-22	4	900.00	R18	2012-09-10	S01
BL000019	SQL Fundamentals	Programming	Manoj Kumar	Pearson	2012-07-28	3	350.00	R19	2012-10-01	S02
BL000020	Internet of Things	Technology	Harish Rao	McGraw Hill	2012-08-30	2	590.00	R20	2012-10-15	S03
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Sample Data — LMS Book Issue

LMS Book Issue

BOOK_ISSUE_NO	MEMBER_ID	BOOK_CODE	DATE_ISSUE	DATE_RETURN	DATE_RETURNED	BOOK_ISSUE_STATUS	FINE_RANGE
1	M001	BL000001	2012-03-20	2012-04-03	2012-04-02	Y	F01
2	M002	BL000002	2012-04-01	2012-04-15	NULL	N	F01
3	M003	BL000003	2012-04-01	2012-04-15	2012-04-12	Y	F01
4	M004	BL000004	2012-04-01	2012-04-15	NULL	N	F02
5	M005	BL000005	2012-04-05	2012-04-19	2012-04-18	Y	F01
6	M006	BL000006	2012-04-10	2012-04-24	NULL	N	F01
7	M007	BL000007	2012-04-12	2012-04-26	2012-04-25	Y	F01
8	M008	BL000008	2012-04-15	2012-04-29	NULL	N	F03
9	M009	BL000009	2012-04-20	2012-05-04	2012-05-02	Y	F01
10	M010	BL000010	2012-04-25	2012-05-09	NULL	N	F01
11	M011	BL000011	2012-05-01	2012-05-15	2012-05-14	Y	F01
12	M012	BL000012	2012-05-05	2012-05-19	NULL	N	F02
13	M013	BL000013	2012-05-10	2012-05-24	2012-05-22	Y	F01
14	M014	BL000014	2012-05-15	2012-05-29	NULL	N	F01
15	M015	BL000015	2012-05-20	2012-06-03	2012-06-01	Y	F01
16	M016	BL000016	2012-05-25	2012-06-08	NULL	N	F03
17	M017	BL000017	2012-06-01	2012-06-15	2012-06-14	Y	F01
18	M018	BL000018	2012-06-05	2012-06-19	NULL	N	F01
19	M019	BL000019	2012-06-10	2012-06-24	2012-06-23	Y	F01
20	M020	BL000020	2012-06-15	2012-06-29	NULL	N	F01
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Relationships & Keys

Primary keys identify records; foreign keys connect related tables

Members → Book Issue

LMS_MEMBERS.MEMBER_ID is referenced by LMS_BOOK_ISSUE.MEMBER_ID.

Books → Book Issue

LMS_BOOK_DETAILS.BOOK_CODE is referenced by LMS_BOOK_ISSUE.BOOK_CODE.

Fine Details → Book Issue

LMS_FINE_DETAILS.FINE_RANGE is referenced by LMS_BOOK_ISSUE.FINE_RANGE.

Suppliers → Book Details

LMS_SUPPLIERS_DETAILS.SUPPLIER_ID is referenced by LMS_BOOK_DETAILS.SUPPLIER_ID.

Design note

A member can have multiple issue records, and a book can appear in multiple issue records over time.

Sample SQL Queries

Examples of questions the database can answer

Permanent members

```
SELECT MEMBER_ID, MEMBER_NAME, CITY
FROM LMS_MEMBERS
WHERE MEMBERSHIP_STATUS = 'Permanent';
```

Books by authors starting with P

```
SELECT BOOK_CODE, BOOK_TITLE,
AUTHOR
FROM LMS_BOOK_DETAILS
WHERE AUTHOR LIKE 'P%';
```

Books by category

```
SELECT CATEGORY, COUNT(*) AS
NO_OF_BOOKS
FROM LMS_BOOK_DETAILS
GROUP BY CATEGORY;
```

Members with unreturned books

```
SELECT DISTINCT M.MEMBER_ID,
M.MEMBER_NAME
FROM LMS_MEMBERS M JOIN
LMS_BOOK_ISSUE I
ON M.MEMBER_ID = I.MEMBER_ID
WHERE I.BOOK_ISSUE_STATUS = 'N';
```

Conclusion

- To conclude, this Library Management System SQL project helped me gain practical knowledge of retrieving and managing data using SQL queries. I used concepts such as SELECT, WHERE, JOIN, LEFT JOIN, LIKE, COUNT(), GROUP BY, aliases, and date filtering to retrieve meaningful information about library members, books, book issues, and membership details.
- These 10 queries demonstrate how SQL can be used to filter records, identify books that have not been returned, count books by category and publisher, and retrieve information based on specific conditions.
- Overall, this project strengthened my understanding of SQL concepts and their practical applications in database management.

Thank you.